

**Request for Information
DARPA-SN-26-33**

**Defense Advanced Research Projects Agency
Tactical Technology Office**

Responses due 15 May2026, by 4:00 PM Eastern Time (ET).

This Request for Information (RFI) from the Defense Advanced Research Projects Agency (DARPA)'s Tactical Technology Office (TTO) seeks information regarding capabilities for the following two categories.

- 1) Unmanned aerial vehicle (herein referred to as “drones”) in the Group 1-3 space with capabilities for fully autonomous launch, recovery, storage, organization, recharging/refueling, organization, internal logistics management, and pre/post-flight checkout. Proposed drone designs must form a mission-focused, collaborative constellation. Responses must be cognizant of long endurance drone constellations with high operational availability and constellation management.

When responding to this category, use Attachment 1 and follow instructions provided in the *Requested Information* section below.

- 2) Storage containers (herein referred to as “containers”) that provide fully autonomous drone storage, logistics management, launch, recovery, and recharge/refuel, while conforming to the intention of a standardized military container (e.g. Conex, 463L pallets, Tricon, ISU container, etc.). Innovative ideas and non-standard containers (e.g. suitcase-based distributed systems, box-based systems) will also be considered within the context of the presented approach, but solutions should be compatible with current military transport capabilities. It is envisioned that these containers shall be self-sufficient with consideration of energy storage, communication equipment, and compute capability.
When responding to this category use Attachment 2 and follow instructions provided in the *Requested Information* section below.

Participants may respond to either one or both categories described above using separate Attachments (provided).

RFI Responses may be used as a market analysis tool informing DARPA about potential vendors for a new program seeking to advance fully autonomous and containerized, deployment, sustainment, and operational support of unmanned autonomous platforms in the Group 1-3 space.

BACKGROUND:

Existing commercial, airborne Group 1-3 platforms are limited in endurance, payload capacity, and onboard electrical auxiliary power. When operated as constellations, they

typically require substantial infrastructure and basing area for deployment and recovery. These constellations typically require human involvement to recover, recharge/refuel, and launch again, lacking full autonomy necessary to achieve sustained operations spanning days or longer. The landscape of current platform technologies has broad limitations that require evolution to achieve high-endurance constellations consisting of drones with meaningful payload Size Weight, Power, and Cost (SWaP-C) staged from fully autonomous containers capable of complete mission-cycle management inclusive of launch, sustainment/swap-out, and recovery.

Example limitations include, but are not limited to:

- Drones
 - Electric-based platforms have limited battery discharge depth and recharge cycle limits that restrict flight time and sortie numbers before substantial performance degradation manifests or battery-changes are required.
 - Hybrid-electric solutions add weight and the complexity of onboard fuel-storage, combustion, electric-machines, and high-voltage power transfer.
 - Buoyant solutions often lack fine six degrees of freedom (6DOF) control and typically have lower onboard auxiliary electrical power available without the use of heavy batteries or large solar panels.
 - Fixed wing platforms typically prohibit compact, autonomous launch/recovery solutions without requiring human intervention at some step and lack the ability to hold fixed spatial positions (not required but is a limitation of fixed wing).
- Containers
 - Existing containers often rely upon a Differential Global Positioning System (DGPS) and cannot position themselves in GPS-denied environments.
 - Deployable containers often lack internal volume allocations for communication equipment and fuel-storage necessary to sustain a constellation for multi-day continuous missions.
 - While some constellation delivery solutions can unleash hundreds of drones in under a minute, these platforms often lack the ability to integrate external payloads, with their additional SWaP, that enable meaningful effects/surveillance.
 - Some container solutions provide internal volume sufficient for drones with meaningful payload-size but may lack the deployment and recovery tempo required to rapidly establish a persistent drone constellation of adequate density for the desired effect.
 - Modern containers optimize designs around battery-powered drones (typically low range, low endurance) without payloads and are incapable

of supporting alternative fuel-sourced platforms providing extended flight times necessary to achieve mission relevance.

While sufficient for preplanned lightshows and commercial activities, further advancements are required to exploit constellation technologies in a high platform density and rapid deployment scenario. Full autonomy in launch, mission, recovery, post/pre-flight and re-launch in a continuous operational cycle for extended hours/days is desired.

DARPA has identified an exigent need for highly deployable, versatile-SWaP Group 1-3 platforms, operating in autonomous constellations that are stored within, deployed from, recovered in, and managed by a fully autonomous container, to support a variety of payloads and missions in GPS-denied environments. Advancements in low-SWaP technologies enable constellations comprising a variety of novel payloads, each requiring dedicated power and weight, but capable of operating in synchrony across the constellation. Constellation populations may comprise up to 500 platforms (number may vary as a function of payload type). Each platform will be equipped with a subsystem or independent payload system with the potential to achieve high operational availability for the combined system over multiple-day periods.

DARPA is looking for responses for U.S. manufactured and assembled drones in the Group 1-3 space and containers spanning current and future technologies to chart the mosaic of possibility in advancing autonomous constellation dynamics for the military. Novel configurations that enable multi-day continuous operations with their corollary constellation management software (ideally with path optimization and collision deconfliction) and innovative configurations of autonomous container-based deployment solutions are of particular interest to DARPA.

Forthcoming technology development programs may leverage these RFI results to inform a new program.

A mission-adjacent use-case involves a transportable container deployed to an austere, GPS-denied environment in suboptimal terrain or onboard a host platform that, upon remote operator inputs, activates and autonomously deploys a series of platforms launching from a single area (possibly serially). The platforms traverse some radius away, loiter on-station with an active payload (not all necessarily in the same location nor same payloads), and execute a mission before returning back to the launch area, autonomously recovering, recharging/refueling, and launching again to repeat the mission (without human intervention). The platform quantity is determined by the particular effect, target, and mission. Thus, “constellation” may comprise any number above 2.

Container size is limited by transportability, endurance, storage, and operability objectives.

DESCRIPTION:

This RFI is looking for information on novel, U.S. manufactured and assembled, aerial platforms and containers, and performers that fundamentally enable:

- Autonomy Level 4 (no human participation required except for mission definition) for the system's operation:
 - o E.g., the operator issues a command, then the drones/constellation maintain level 4 autonomy
- Long-constellation endurance (on the order of multiple days)
- Drones with continuous electrical power available to payloads, low-cost Commercial-Off-the-Shelf (COTS) construction, and sized for containerized storage
- Constellation management software (path optimization, collision avoidance planning, constellation reshaping/reorientation, formation flying)
- Autonomous container operation (inclusive but not limited to storage, launch, recovery, recharge/refuel, constellation communications, compute capability, multi-day fuel/energy storage, and constellation management software)

Responders are requested to provide tabular information on either existing or notional platforms that may satisfy this RFI in the attachments provided.

REQUESTED INFORMATION:

DARPA is interested in responses that provide information and insights related to core capability metrics. Response instructions are below.

If responding ONLY to the “Drone” portion of the RFI, see RFI Attachment #1 and complete the Cover Page, Required Data Table, enter any optional Graphics, and use the Technical Description sections as required, entering any References on the last page. Use Attachment #1 to capture all information for your response, keeping all responses to 5 pages maximum (including references). Do not modify Attachment #1's layout or add pages to Attachment #1. Keep all responses to the maximum 5-page limit of Attachment #1.

If responding ONLY to the “Container” portion of the RFI, see RFI Attachment #2 and complete the Cover Page, Required Data Table, enter any optional Graphics, and use the Technical Description sections as required, entering any References on the last page. Use Attachment #2 to capture all information for your response, keeping all responses to 5 pages maximum (including references). Do not modify Attachment #2's layout or add

pages to Attachment #2. Keep all responses to the maximum 5-page limit of Attachment #2.

If responding to both the “Drone” and “Container” portions of the RFI, complete both Attachment #1 and Attachment #2, per the respective attachment instructions, and submit both documents together.

Responders are encouraged to provide information about an existing, exemplary system that fits this space, if available. If none currently exists, responders are recommended to size appropriate systems based on their experience and abilities to populate information within the requested table below. The opportunity to list up to 2 bounding examples is provided within the tabular body of both Attachment #1 and Attachment #2 (up to 2 examples for each attachment). Examples may be existing, proposed, or conceptual.

SUBMISSION:

This announcement contains all information required to submit a response. No additional forms, kits, or other materials are needed.

Submissions in response to the RFI should be concise. Responders should submit a single integrated response addressing the areas described above. DARPA will only review submissions received as a Microsoft Word (.doc or .docx) file or unprotected Adobe Acrobat (.pdf) file. Each response is limited to the 5-page limit of each Attachment #1 and Attachment #2 (total of up to 10 pages if both Attachments are submitted) using 12-point font and 1-inch margins on 8.5-inch by 11-inch paper. Effective responses that can be provided in fewer than 5 pages are encouraged.

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Attachment #1 and Attachment #2 include a template all responders are required to complete for their submission.

SUBMISSION INSTRUCTIONS AND CONTACT INFORMATION

All unclassified responses to this RFI must be emailed to DARPA-SN-26-33@darpa.mil. Responses will be accepted until 4 p.m. ET on 15 May, 2026. Early responses are encouraged.

DO NOT SUBMIT CLASSIFIED INFORMATION TO THIS ADDRESS.

Please see Attachment 3 – Security to this notice for instructions on submitting classified information.

All technical and administrative correspondence and questions regarding this RFI must be sent to the same email address. Emails pertaining to this RFI that are sent directly to individual DARPA program managers will not receive a response. No telephone inquiries will be accepted.

NOTE: DARPA may conduct discussions with individual respondents as necessary to gain a full understanding of the information submitted.

DISCLAIMERS AND IMPORTANT NOTES:

- This is an RFI issued solely for information and new program planning purposes; it does not constitute a formal solicitation for proposals or proposal abstracts. In accordance with FAR 15.201(e), responses to this notice are not offers and cannot be accepted by the Government as such.
- DARPA will not provide reimbursement for costs incurred in responding to this RFI.
- If proprietary information is submitted, it must be appropriately and specifically marked. It is the submitter's responsibility to clearly define to the Government what is considered proprietary data. Any proprietary information should clearly be labeled as "Proprietary." DARPA will not publicly disclose proprietary information obtained as a result of the RFI. To the full extent that it is protected pursuant to the Freedom of Information Act and other laws and regulations, information identified by a respondent as "Proprietary" will be appropriately controlled. Responses to this RFI will not be returned.
- Respondents are advised that DARPA is under no obligation to acknowledge receipt of the information received or provide feedback to respondents with respect to any information submitted under this RFI. Responses to this RFI do not bind DARPA to any further actions.
- DARPA will disclose submission contents only for the purpose of review. Submissions may be reviewed by the Government (DARPA and other Government partners), Federally Funded Research and Development Centers (FFRDCs), and Scientific, Engineering and Technical Assistance (SETA) support contractors bound by appropriate non-disclosure agreements.